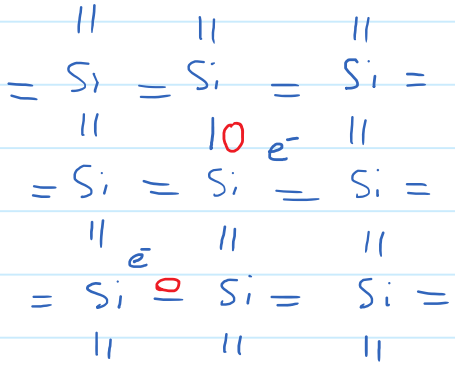
Reading

Adel S. Sedra and Kenneth C. Smith, **Microelectronic Circuits** 7th Edition, *Oxford University Press*, 2014.

- Chapter 3.4-3.6

Review

$n_p = n_e$



T ↓ 10K T ↑

$n_i = B T^{3/2} e^{-E_g/2kT}$ (E_g = 1.12 eV)

$n_i = n_e = n_p$

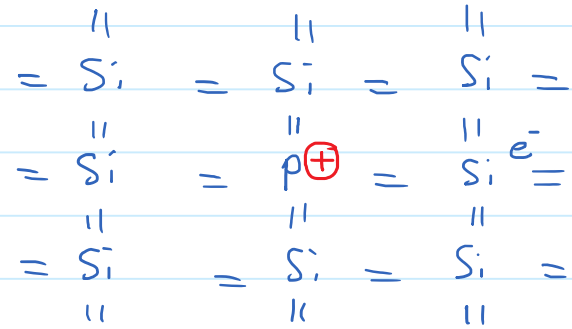
@ 300 K → $n_i = 1.5 \times 10^{10}$ carriers/cm³

$$n_p . n_e = n_i^2$$

4A 5A

Si → P

n-type semiconductor



$n_e = N_D$ $n_p . n_e = n_i^2$

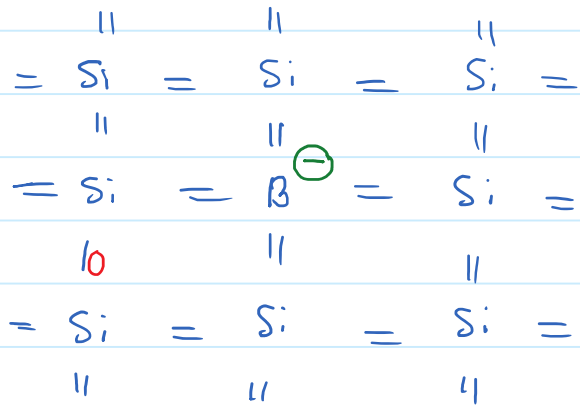
$n_p \cdot N_D = n_i^2$

$$n_p = \frac{n_i^2}{N_D}$$

4A 3A

Si → B

p-type semiconductor

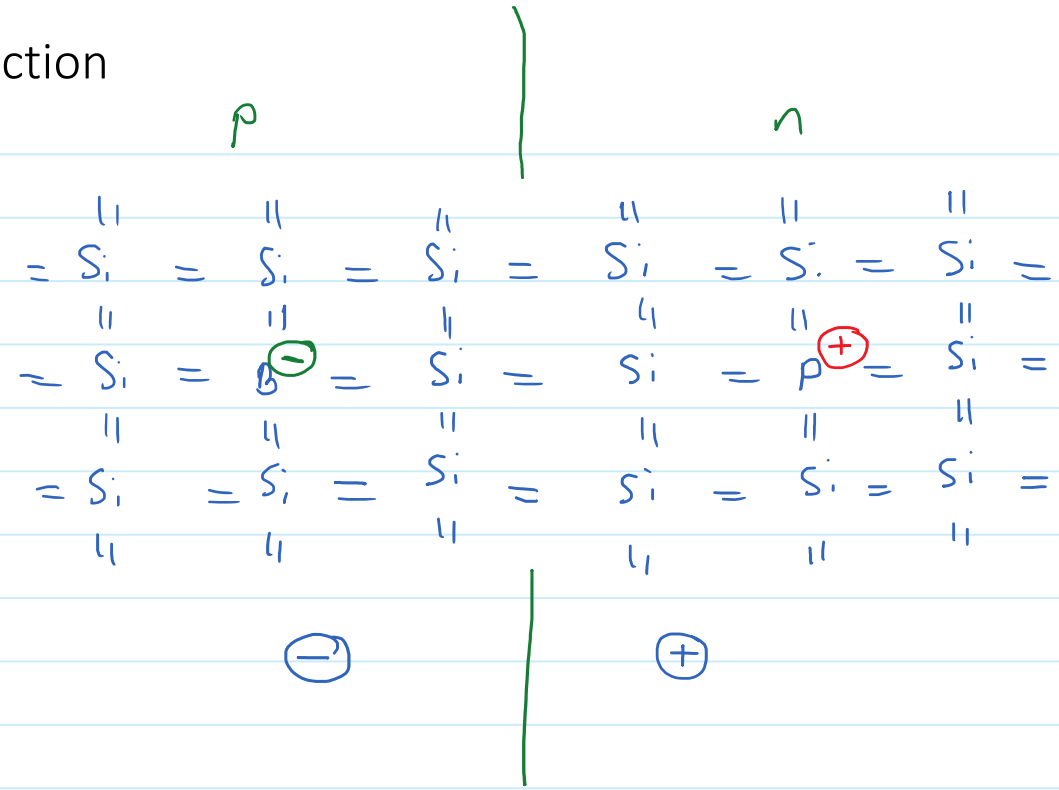


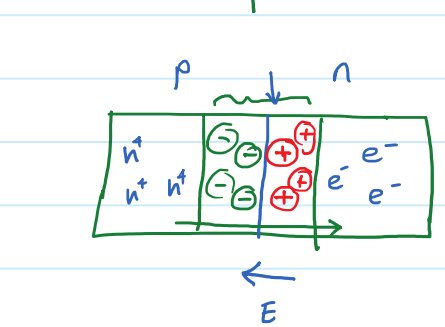
$n_p = N_A$ $n_e = ?$

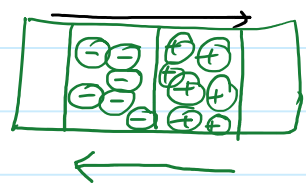
$n_i^2 = n_p n_e = N_A . n_e$

$$n_e = \frac{n_i^2}{N_A}$$

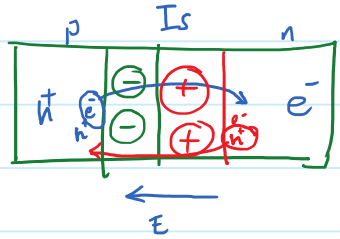
pn Junction

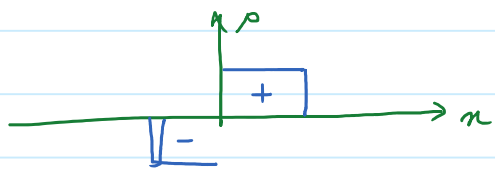




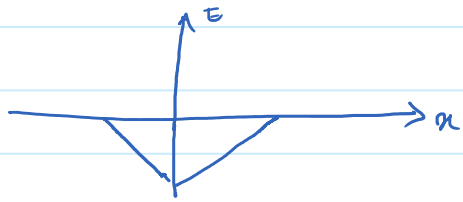




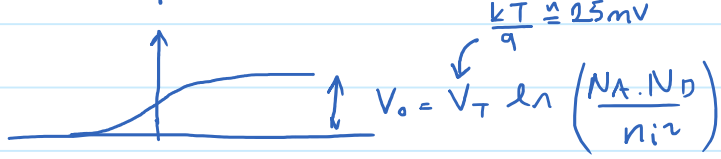




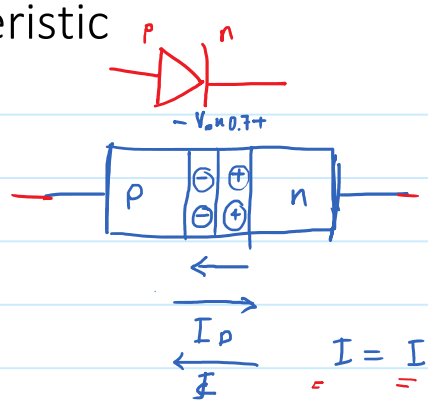
$$\frac{dE}{dx} = \nabla . E = \frac{\rho}{\varepsilon}$$



$$E = -\nabla V$$



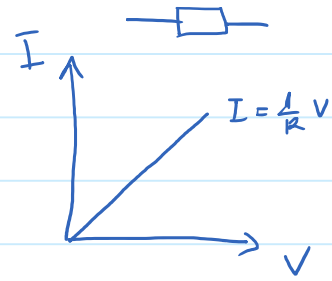
I-V Characteristic

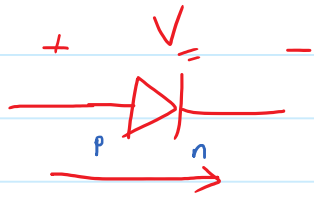


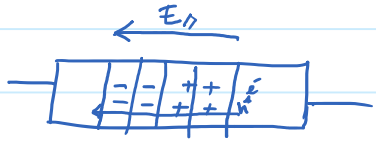
I_D

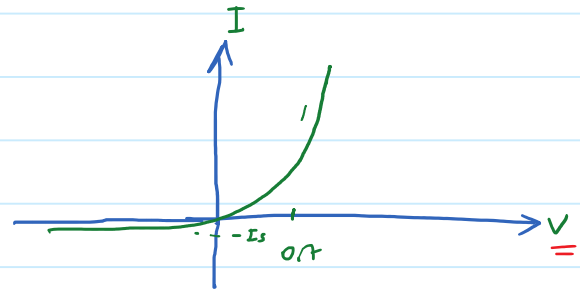
I

$I = I_D - I_S$









$$I = I_S \left(e^{V/V_T} - 1 \right)$$

25mV

Reverse Breakdown

